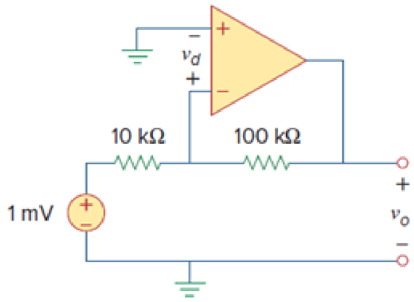
Problem

The op amp in Fig. 5.46 has Ri = 100 kΩ, Ro = 100 Ω, A = 100,000. Find the differential voltage vd and the output voltage vo.

Figure. 5.46:



Step-by-step solution

Step 1 of 5

Refer to circuit diagram in Figure 5.46 in the textbook.

Draw the equivalent circuit of the circuit shown in Figure 5.46.

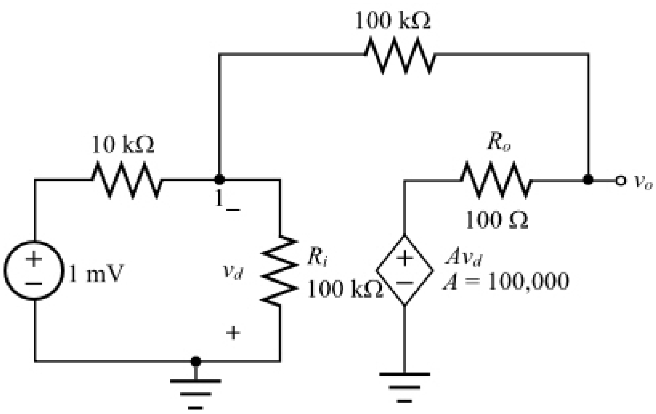


Figure 1

Step 2 of 5

From the circuit shown in Figure 1,

$$v_1 = -v_d$$

Apply Kirchhoff's current law at node 1.

$$\frac{v_1 - 1\text{ mV}}{10\text{ k}\Omega} + \frac{v_1}{100\text{ k}\Omega} + \frac{v_1 - v_o}{100\text{ k}\Omega} = 0$$

$$\frac{v_1}{10\text{ k}\Omega} + \frac{v_1}{100\text{ k}\Omega} + \frac{v_1}{100\text{ k}\Omega} - \frac{v_o}{100\text{ k}\Omega} = \frac{1\text{ mV}}{10\text{ k}\Omega}$$

$$\left(\frac{1}{10\text{ k}\Omega} + \frac{1}{100\text{ k}\Omega} + \frac{1}{100\text{ k}\Omega}\right)v_1 - \frac{v_o}{100\text{ k}\Omega} = \frac{1\text{ mV}}{10\text{ k}\Omega}$$

$$\left(\frac{10+1+1}{100\text{ k}\Omega}\right)v_1 - \frac{v_o}{100\text{ k}\Omega} = \frac{1\text{ mV}}{10\text{ k}\Omega}$$

$$12v_1 - v_o = 10\text{ mV}$$

Substitute $-v_d$ for v_1 in the equation.

$$-12v_d - v_o = 10\text{ mV} \quad \text{...... (1)}$$

Step 3 of 5

Apply Kirchhoff's current law at node v_o.

$$\frac{v_o - v_1}{100\text{ k}\Omega} + \frac{v_o - Av_d}{100\ \Omega} = 0$$

Substitute $-v_d$ for v_1 in the equation.

$$\frac{v_o + v_d}{100\text{ k}\Omega} + \frac{v_o - Av_d}{100\ \Omega} = 0$$

$$\frac{v_o}{100\text{ k}\Omega} + \frac{v_o}{100\ \Omega} + \frac{v_d}{100\text{ k}\Omega} - \frac{100000v_d}{100\ \Omega} = 0 \qquad (A = 100{,}000)$$

$$\left(\frac{1}{100\text{ k}\Omega} + \frac{1}{100\ \Omega}\right)v_o + \left(\frac{1}{100\text{ k}\Omega} - \frac{100000}{100\ \Omega}\right)v_d = 0$$

$$\left(\frac{1+1000}{100\text{ k}\Omega}\right)v_o + \left(\frac{1-10^8}{100\text{ k}\Omega}\right)v_d = 0$$

$$1001v_o - 99999999v_d = 0$$

$$1001v_o = 99999999v_d$$

$$v_o = 99900.099v_d \quad \text{...... (2)}$$

Step 4 of 5

Recall equation (1).

$$-12v_d - v_o = 10\text{ mV}$$

Substitute $99900.099v_d$ for v_o in the equation.

$$-12v_d - 99900.099v_d = 10\text{ mV}$$

$$-99912.099v_d = 10\text{ mV}$$

$$v_d = -100\text{ nV}$$

Therefore, the differential voltage, v_d is $\boxed{-100\text{ nV}}$.

Step 5 of 5

Calculate the output voltage.

Recall equation (2).

$$v_o = 99900.099 v_d$$

Substitute $-100\ \text{nV}$ for v_d in the equation.

$$\begin{aligned} v_o &= (99900.099)(-100\ \text{nV}) \\ &= -9.99\ \text{mV} \\ &= -10\ \text{mV} \end{aligned}$$

Therefore, the output voltage, v_o is $\boxed{-10\ \text{mV}}$.